National Institute Of Technology | 2025

AUTONOMOUS CAR FOR DEFENCE SECURITY SURVEILLANCE

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Date : 19 March 2025

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PROJECT OVERVIEW

- This project aims to design an autonomous vehicle that can navigate its environment without human intervention, specifically targeted for applications like border surveillance and patrolling.

- Due to resource and time constraints, the full surveillance model could not be built. Instead, a working obstacle-avoiding car prototype was implemented using Arduino Uno, ultrasonic sensors, a servo motor, and a motor driver module.

- The prototype showcases the basic concept of autonomous navigation and sets the foundation for future enhancement toward a fully autonomous border surveillance system.

MOTIVATION

- Border areas are often difficult to access, dangerous, and vast, making constant human patrolling challenging.
- An autonomous surveillance vehicle can provide 24/7 monitoring, reduce human risk, and increase efficiency.
- This inspired the idea to design a vehicle that can sense obstacles, make decisions, and move autonomously.
- The current project represents the first step toward achieving that goal using affordable and easily available components.

INTRODUCTION

What is an Autonomous Vehicle?

- A vehicle that can navigate without human intervention using sensors, AI algorithms, and embedded systems.
- Used in various fields: transportation, defense, industrial automation, and surveillance.

Importance in Border Security

- Provides unmanned surveillance in hostile terrains.
- Reduces human risk and increases efficiency in patrolling.
- Detects intrusions, tracks movements, and reports anomalies in real-time.

- Security and surveillance are critical needs in sensitive areas like national borders.
- Traditional border patrolling involves risks to human life and is often inefficient.

- Autonomous vehicles offer a safer, cost-effective alternative for continuous monitoring.
- This project focuses on building a prototype autonomous car using Arduino Uno.
- Due to hardware and time constraints, a simplified obstacle-avoiding car was developed.

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- The car uses:
  - Ultrasonic sensor for obstacle detection
  - Servo motor to scan directions
  - L298N motor driver to control motion
- It makes decisions based on sensor data and navigates without manual input.
- Demonstrates the basic concept of autonomous navigation.
- Serves as a stepping stone for future development of AI-powered surveillance vehicles.

LITERATURE REVIEW (RELATED WORK)

Past Research on Autonomous Vehicles

- Zhang et al. (2022): Used sensor fusion (LIDAR, cameras, GPS) for navigation in unknown terrains.
- Kumar et al. (2023): Analyzed Unmanned Ground Vehicles (UGVs) for border security.
- Smith et al. (2020): Implemented deep learning-based sensor fusion for autonomous navigation.
- Brown et al. (2020): Developed AI-driven decision-making models for real-time vehicle control.

Limitations in Existing Systems

- High Cost: Most systems rely on expensive LIDAR technology.
- Navigation Challenges: AI models require real-time adaptability in dynamic environments.
- Limited Low-Power Solutions: ESP32-based navigation is not widely explored in security applications.

RESEARCH GAP

- Most current border surveillance systems rely heavily on manual patrolling, risking human lives in hazardous and remote terrains.
- Fully autonomous surveillance vehicles with real-time decision-making, obstacle avoidance, and live monitoring are still in early research or high-cost development stages.
- Existing obstacle-avoiding robots are often limited to indoor or structured environments and lack scalability for rough outdoor terrains.
- Many low-cost robotic prototypes do not integrate GPS, vision systems, or AI, limiting their functionality for real-world surveillance applications.
- There is a lack of affordable, modular, and field-deployable prototypes that combine obstacle detection with surveillance capabilities.
- Research in autonomous mobility for border regions often lacks emphasis on energy efficiency, real-time communication, and terrain adaptability.
- Most available models do not utilize multi-sensor data fusion (ultrasonic, camera, GPS, gyroscope) to enhance reliability in dynamic environments.

OBJECTIVES

This project aims to:

- Develop an autonomous ground vehicle using Arduino uno and multiple sensors.
- To design and develop a basic obstacle-avoiding autonomous car using Arduino Uno.
- To implement real-time distance measurement using an ultrasonic sensor.
- To use a servo motor to enable directional scanning (left, front, right) for obstacle detection.
- To control car movement (forward, left, right) through the L298N motor driver module.
- To implement simple decision-making logic based on sensor input to avoid collisions.
- To demonstrate autonomous navigation capabilities in a controlled indoor environment.
- To validate the core principles of autonomy and apply them as a foundation for future enhancements.
- To explore the feasibility of scaling this prototype into a fully autonomous vehicle for border surveillance.
- Enhance border security patrols through automation.

METHODOLOGY

Component Selection

- Chose low-cost and accessible components: Arduino Uno, HC-SR04 ultrasonic sensor, SG90 servo motor, L298N motor driver, and DC motors.

System Design

- Mounted the ultrasonic sensor on the servo motor to allow scanning in three directions (left, front, right).
- Connected components according to circuit requirements using jumper wires and a breadboard.

Programming & Logic Implementation

- Coded in Arduino IDE to control the servo motor and read distance from the ultrasonic sensor.
- Programmed logic to compare distance readings and decide the safest direction to move.
- Used digitalWrite() and analogWrite() to control motor direction and speed via L298N.

Loop Execution

- Created a continuous loop using loop() to scan surroundings and take real-time navigation decisions.
- Incorporated delays to stabilize sensor readings and avoid jittery movements.

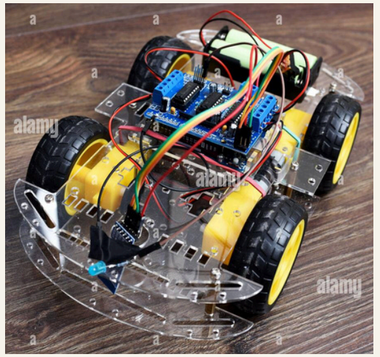
Testing & Debugging

- Tested the prototype in various indoor layouts with multiple obstacle placements.
- Adjusted scanning angles and distance thresholds to improve obstacle avoidance performance.

Performance Evaluation

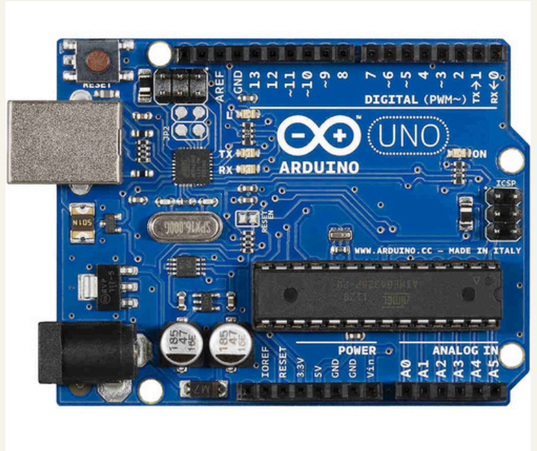
- Evaluated the car's ability to avoid collisions and make decisions autonomously in a simple environment.

COMPONENTS



FRAME

ULTRASONIC SENSOR

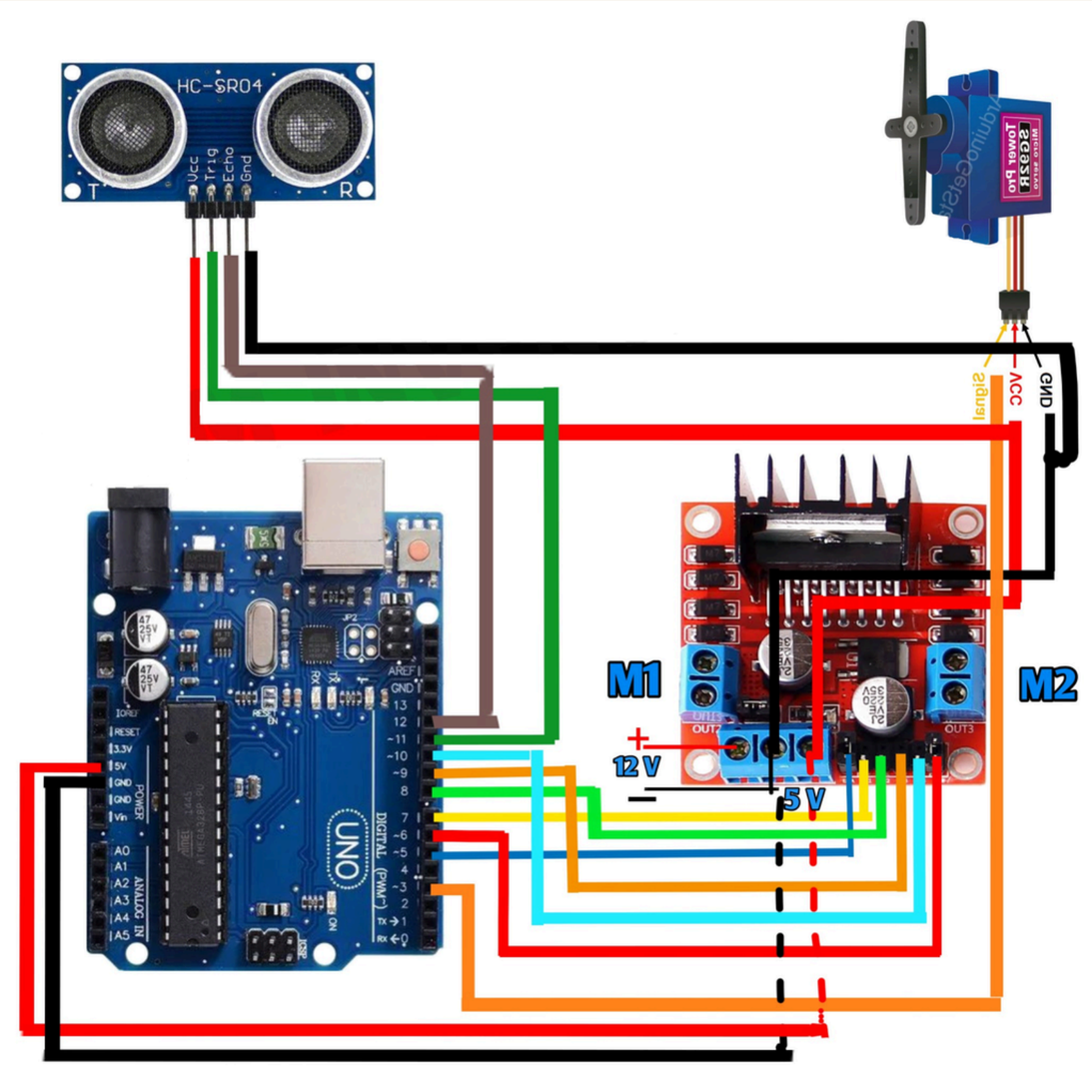


ARDUINO UNO

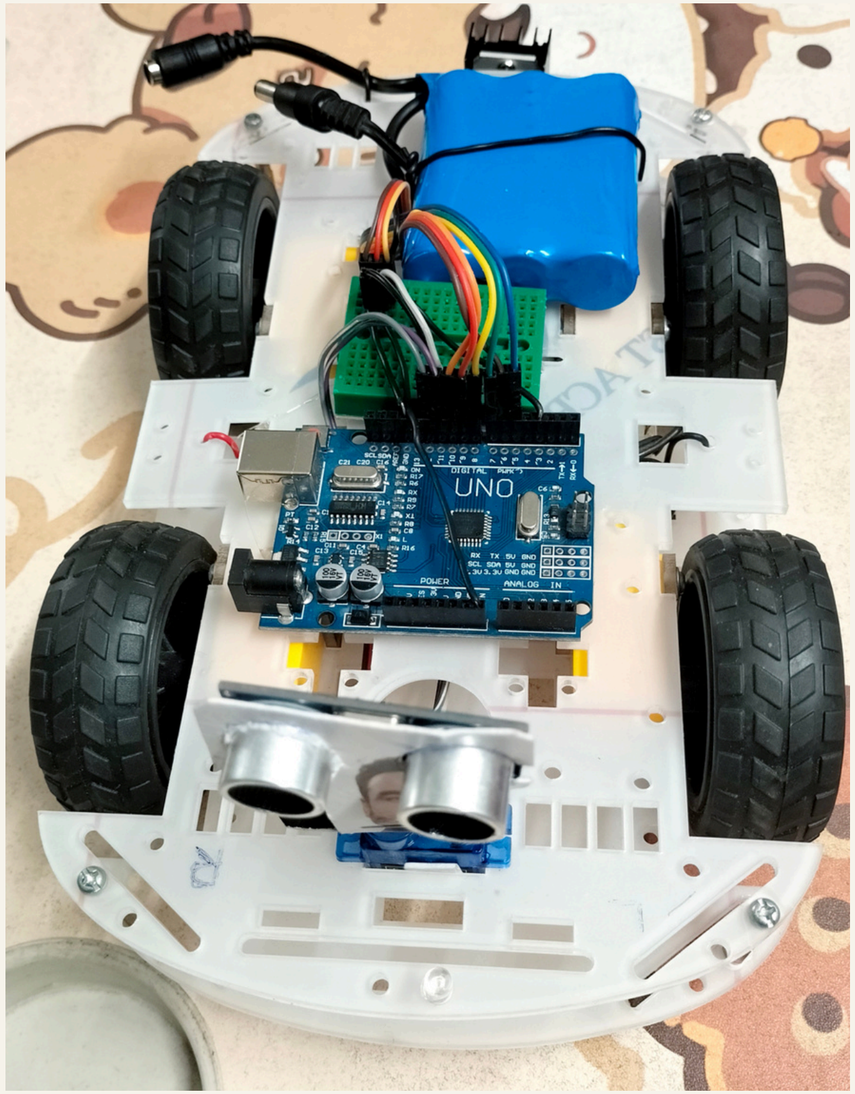
BO MOTOR

SERVO MOTOR

CIRCUIT DIAGRAM



PROTOTYPE



SYSTEM ARCHITECTURE

- The ultrasonic sensor is mounted on the servo motor to enable scanning in three directions: Left, Front, and Right.
- The Arduino collects distance data from these directions.
- Based on pre-set conditions, the Arduino decides whether to move forward or turn.
- The L298N motor driver receives commands from Arduino to control the motors.
- This loop continues in real-time, allowing the car to avoid obstacles and move autonomously.

WORKING MECHANISM

- The servo motor rotates the ultrasonic sensor to 0° (Left), 90° (Front), and 180° (Right).
- At each position, the sensor records the distance to any obstacle.
- If the front distance is less than a threshold (e.g., 30 cm):
- The car compares left and right distances.
- It turns toward the direction with the greater distance (i.e., clearer path).
- If there's no obstacle in front, the car continues moving forward.
- The process is continuous and handled within the loop() function of the Arduino sketch.

PROGRAMMING LOGIC

- The code initializes all components in setup().
- Servo rotates to different angles; distance values are measured.
- A threshold value (e.g., 30 cm) determines when to turn.
- digitalWrite() is used to control motor movement.
- delay() functions are used briefly to stabilize readings and avoid erratic movement.
- The logic continuously repeats in the loop() for real-time navigation.

RESULTS

- The car was able to detect obstacles in real-time using the ultrasonic sensor.
- It could decide independently whether to stop, move forward, or turn.
- In a controlled indoor environment, the car moved around without collisions.
- Adjustments to delay times, angles, and threshold distances improved performance.
- The car showcased autonomous decision-making and motion control using affordable and beginner-friendly components.

DISCUSSION

- The obstacle-avoiding mechanism worked successfully in indoor, flat-terrain settings.
- It demonstrated basic autonomous vehicle behavior like sensing and decision-making.
- Limitations include:
- No memory of past paths (i.e., no mapping or localization).
- Limited to short-range detection using a single sensor.
- Ineffective in complex outdoor or rugged environments.
- Despite the constraints, the prototype validated the core logic needed for autonomous navigation.

FUTURE SCOPE

To convert this prototype into a fully autonomous surveillance vehicle for border security:

1. GPS and Compass – For real-time location tracking and route navigation.
2. Cameras (ESP32-CAM / Pi Camera) – To capture live video feed for surveillance.
3. Machine Learning Models – For recognizing human figures, intrusions, or objects.
4. Advanced Sensors – Such as IR, LIDAR, radar for better accuracy and range.
5. SLAM Algorithm – For mapping and localization in unknown terrains.
6. Wireless Modules (Wi-Fi/LoRa/GSM) – To send data to a remote control center.
7. All-Terrain Design – To handle outdoor rugged landscapes effectively.

These enhancements will allow the vehicle to patrol autonomously, detect threats, and provide live monitoring to border control units.

IMPLEMENTATION

STEPS:

1. Making of car controlled by remote.
2. Making of vehicle which can move by scanning the obstacles.
3. learning about the different sensors.like-Lora sensor, gyroscope, gps, etc.
4. Making of the autonomous vehicle.

CONCLUSION & FUTURE WORK

- The implemented car serves as a proof of concept for autonomous navigation.
- It successfully avoided obstacles using simple logic and components.
- Though limited in its current form, it lays the foundation for a future autonomous vehicle for border security.
- The project has immense potential to be scaled into a cost-effective, intelligent surveillance system with added capabilities and smarter technologies.

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THANK YOU

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